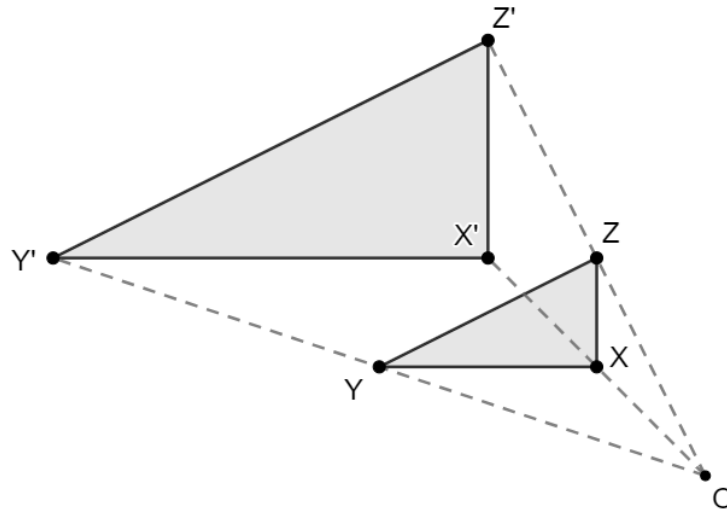
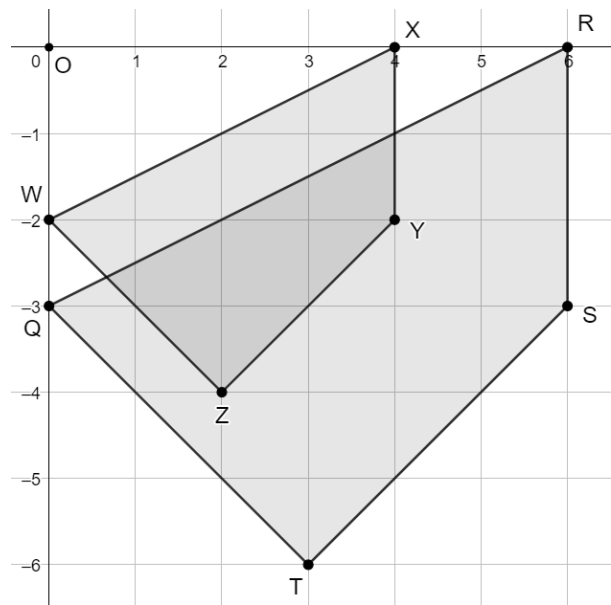


$\triangle X'Y'Z'$ was created by dilating $\triangle XYZ$. The length of $\overline{XY}$ is 2 and the length of $\overline{X'Y'}$ is 4.



- $\triangle XYZ$ is the
☐ image
☐ preimage
 , $\triangle X'Y'Z'$ is the
☐ image
☐ preimage
 , and the center of dilation is point
☐ X
☐ X'
☐ O
.
- Determine the dilation ratio.
- If the length of $\overline{Y'Z'}$ is $2\sqrt{5}$, determine the length of $\overline{YZ}$.
- What is the ratio of $\frac{\overline{XZ}}{\overline{X'Z'}}$?
- What is the ratio of $\frac{\overline{YY'}}{\overline{OY}}$?

$WXYZ$ and $QRST$ are shown on the coordinate grid, where $WXYZ$ is a dilation of $QRST$.



6. Complete the table:

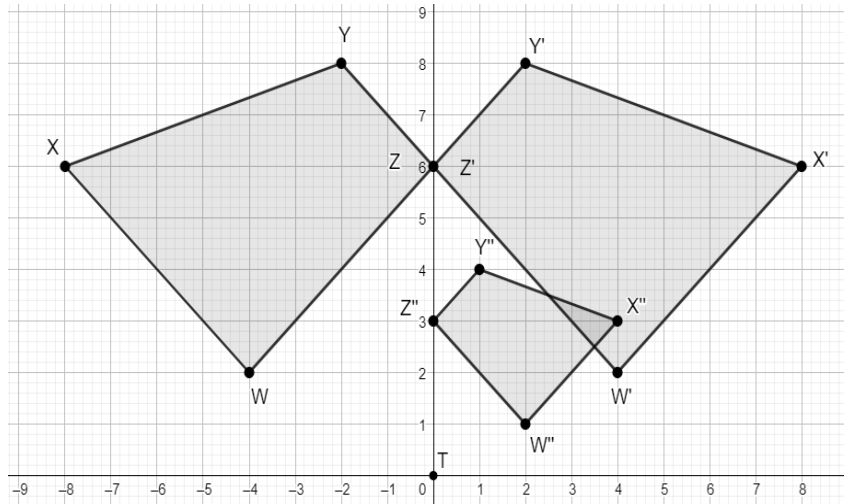
Segment	Slope	Distance
$\overline{YZ}$		
$\overline{ST}$		
$\overline{WX}$		
$\overline{QR}$		

7. Show the dilation on the coordinate grid by drawing dotted lines from the center of dilation through each vertex.
8. Write a description of the dilation by giving the center of dilation and the scale factor.
9. What is the ratio $\frac{\overline{OX}}{\overline{OR}}$?
10. What is the ratio $\frac{\overline{OZ}}{\overline{OT}}$?

Geometry
Unit 6
Lesson 2 Practice

Name _____
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A sequence of transformations of $WXYZ$ is shown on the coordinate grid. $WXYZ$ is reflected then dilated, where $W'X'Y'Z'$ is the reflection and $W''X''Y''Z''$ is the dilation of $W'X'Y'Z'$.



Complete the descriptions:

- $WXYZ$ was _____ to create $W'X'Y'Z'$.
- $W'X'Y'Z'$ was dilated using a scale factor of _____ centered at _____ to create $W''X''Y''Z''$.
- Complete the table:

Transformation	Algebraic Description
Reflection	$(x, y) \rightarrow$
Dilation	$(x, y) \rightarrow$

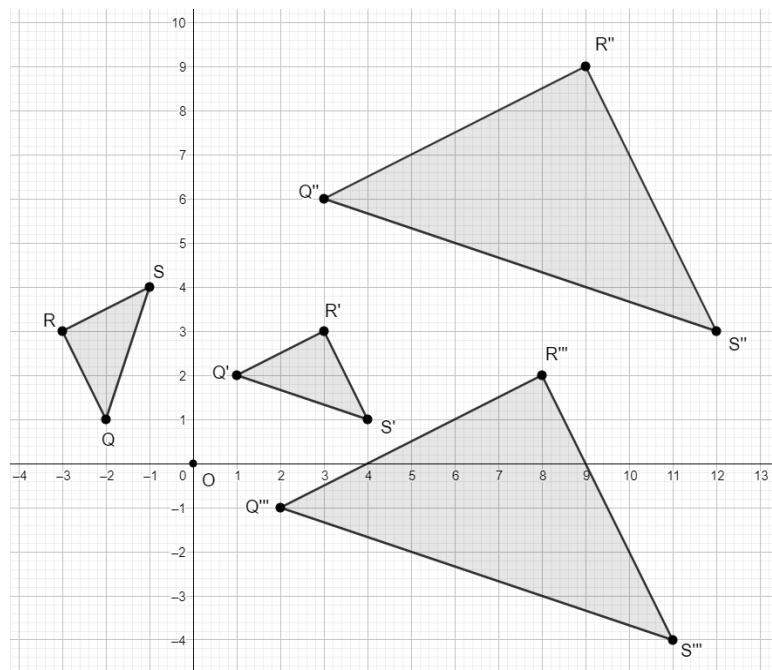
- Determine the coordinates if $WXYZ$ is first dilated then reflected:

Preimage $WXYZ$		Dilation $W'X'Y'Z'$		Reflection $W''X''Y''Z''$	
W	$(-4, 2)$	W'		W''	
X	$(-8, 6)$	X'		X''	
Y	$(-2, 8)$	Y'		Y''	
Z	$(0, 6)$	Z'		Z''	

- Based on the results above, the sequence of transformations commutative.

☐ is
☐ is not

$QRS, Q'R'S', Q''R''S'',$ and $Q'''R'''S'''$ are shown on the coordinate grid.



6. Determine what transformations occurred and in what order:
- QRS is first

☐ translated
☐ reflected
☐ rotated
☐ dilated

 to $Q'R'S'$, then

☐ translated
☐ reflected
☐ rotated
☐ dilated

 to $Q''R''S''$,
- and finally

☐ translated
☐ reflected
☐ rotated
☐ dilated

 to $Q'''R'''S'''$.
7. Describe the transformation for $QRS \rightarrow Q'R'S'$.
8. Describe the transformation for $Q'R'S' \rightarrow Q''R''S''$.
9. Describe the transformation for $Q''R''S'' \rightarrow Q'''R'''S'''$.

10. Write the algebraic description for the sequence of transformations from 7-9:

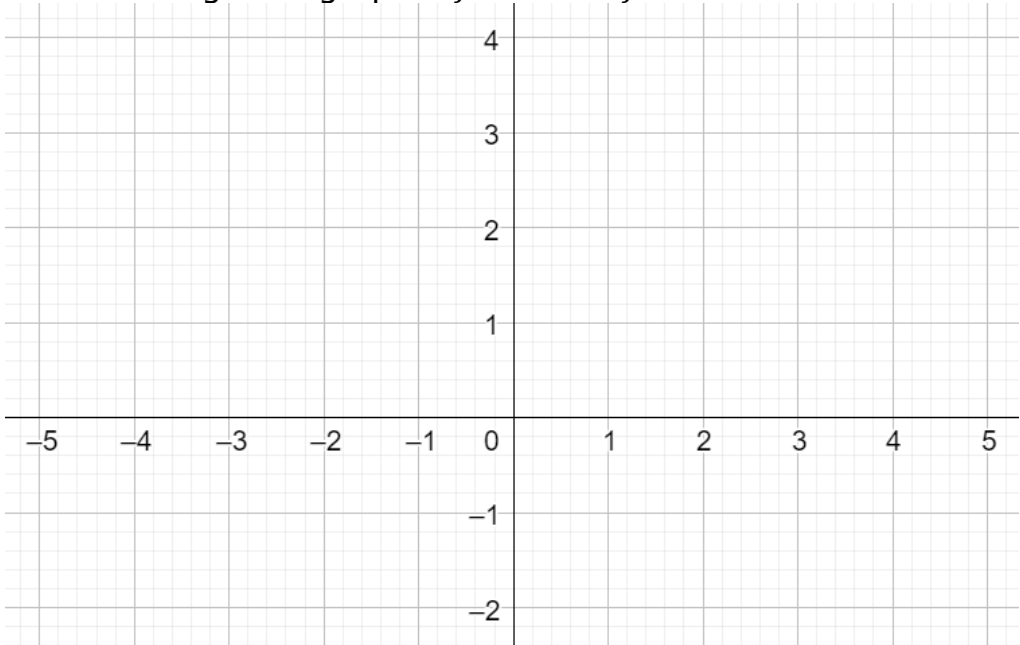
Transformation	Algebraic Description
	$(x,y) \rightarrow$
	$(x,y) \rightarrow$
	$(x,y) \rightarrow$

The coordinates of $\triangle JKL$ and $\triangle J'K'L'$ are shown in the table, where $\triangle JKL$ is the preimage:

$\triangle JKL$		
Vertex	x	y
J	-2	0
K	-3	3
L	-1	2

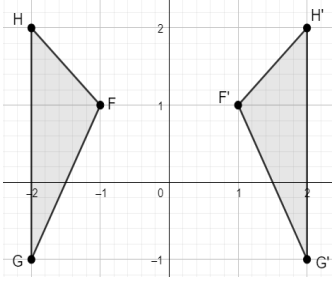
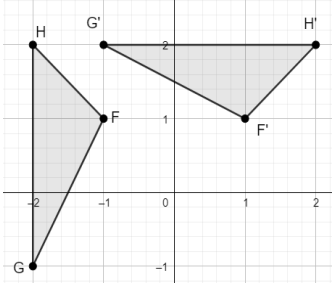
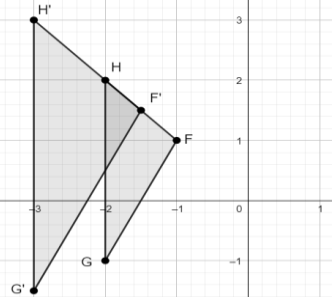
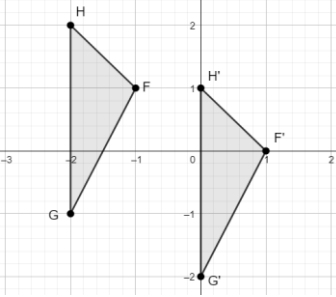
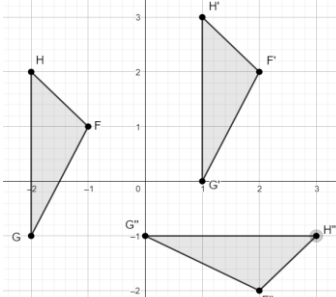
$\triangle J'K'L'$		
Vertex	x	y
J'	2	0
K'	3	3
L'	1	2

1. Use the coordinate grid to graph $\triangle JKL$ and $\triangle J'K'L'$.



2. What type of transformation was applied to the preimage $\triangle JKL$?
3. Was distance preserved in this transformation?
4. $\triangle J'K'L'$ is further transformed by dilating with a scale factor of $\frac{1}{2}$ centered at the origin to create $\triangle J''K''L''$. Graph $\triangle J''K''L''$ on the coordinate grid above.
5. Was distance preserved in this second transformation from $\triangle J'K'L'$ to $\triangle J''K''L''$?

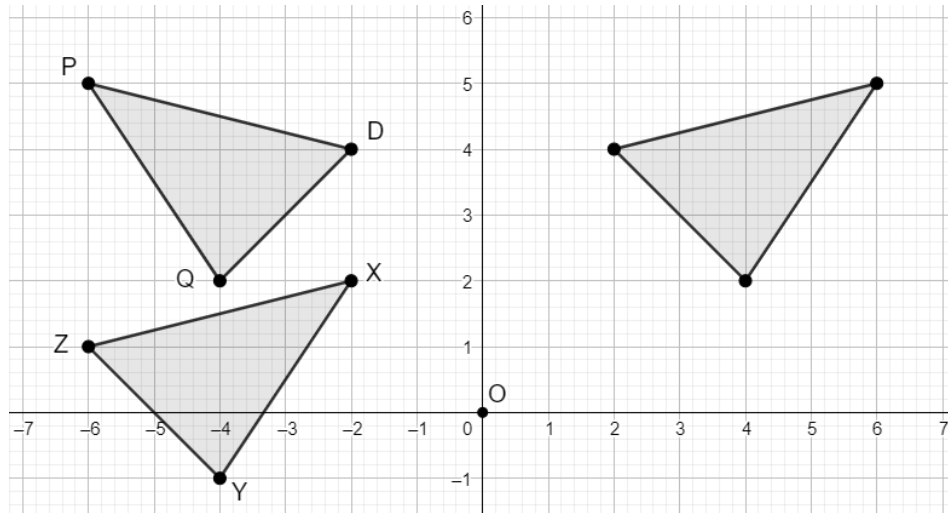
Complete the following statements about the transformation of preimage FGH :

6. Triangle $F'G'H'$ was created by a This transformation	☐ translation. ☐ reflection. ☐ rotation. ☐ dilation. ☐ does ☐ does not preserve distance.	
7. Triangle $F'G'H'$ was created by a This transformation	☐ translation. ☐ reflection. ☐ rotation. ☐ dilation. ☐ does ☐ does not preserve distance.	
8. Triangle $F'G'H'$ was created by a This transformation	☐ translation. ☐ reflection. ☐ rotation. ☐ dilation. ☐ does ☐ does not preserve distance.	
9. Triangle $F'G'H'$ was created by a This transformation	☐ translation. ☐ reflection. ☐ rotation. ☐ dilation. ☐ does ☐ does not preserve distance.	
10. Triangle $F''G''H''$ was created by a transformation sequence of first a ☐ translation. ☐ reflection. ☐ rotation. ☐ dilation. This transformation sequence	☐ translation ☐ reflection ☐ rotation ☐ dilation ☐ does ☐ does not preserve distance.	

Geometry
Unit 6
Lesson 4 Practice

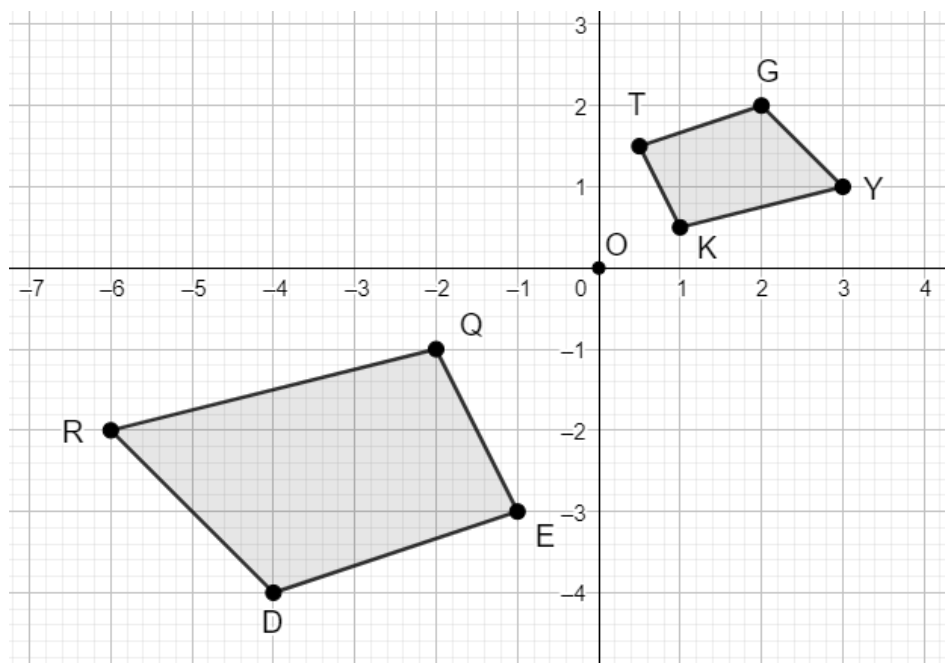
Name _____
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$\triangle XYZ$ is the result of a sequence of transformations of $\triangle PDQ$. The unlabelled triangle is the first transformation.



1. Describe the first transformation of $\triangle PDQ$.
2. Write the transformation algebraically.
3. Describe the second transformation that results in $\triangle XYZ$.
4. Write the second transformation algebraically.
5. Explain if the sequence of transformations preserves distance.

KTGY is the result of a sequence of transformations of QEDR.



Ian and Natalie were determining the sequence of transformations, but each came up with a different sequence. Complete the table for the missing portions of their transformations.

			First Transformation	Second Transformation
6.	Ian's Sequence	Written Description	Rotate 180° about the origin.	
7.		Algebraic Description		$(x,y) \rightarrow (\frac{1}{2}x, \frac{1}{2}y)$
8.	Natalie's Sequence	Written Description	Dilate using a scale factor of $\frac{1}{2}$ centered at the origin.	
9.		Algebraic Description		$(x,y) \rightarrow (-x, -y)$

10. Considering both proposed sequences, you can conclude that

- Ian's sequence is
 - Natalie's sequence is
 - both sequences are
- correct.

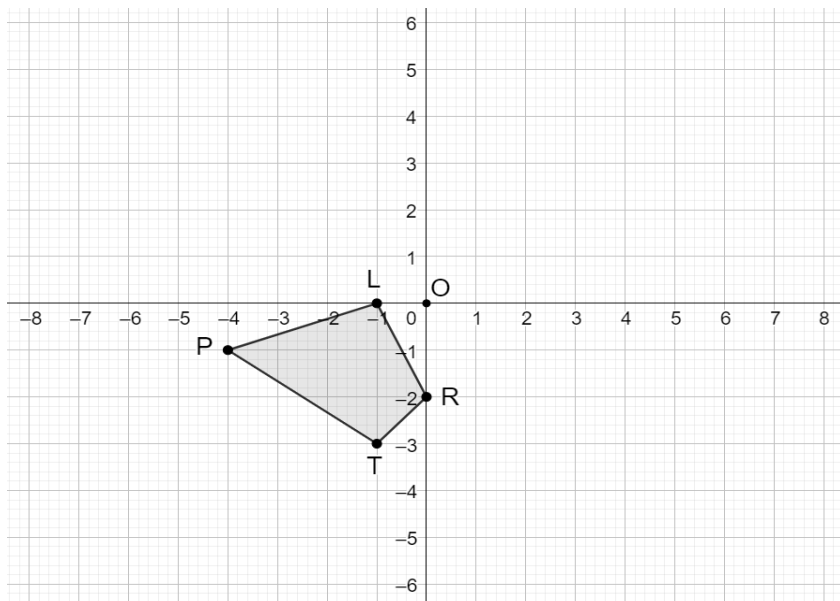
Geometry
Unit 6
Lesson 5 Practice

Name _____
Date _____
Period _____

$L RTP$ and point O are shown on the coordinate plane.

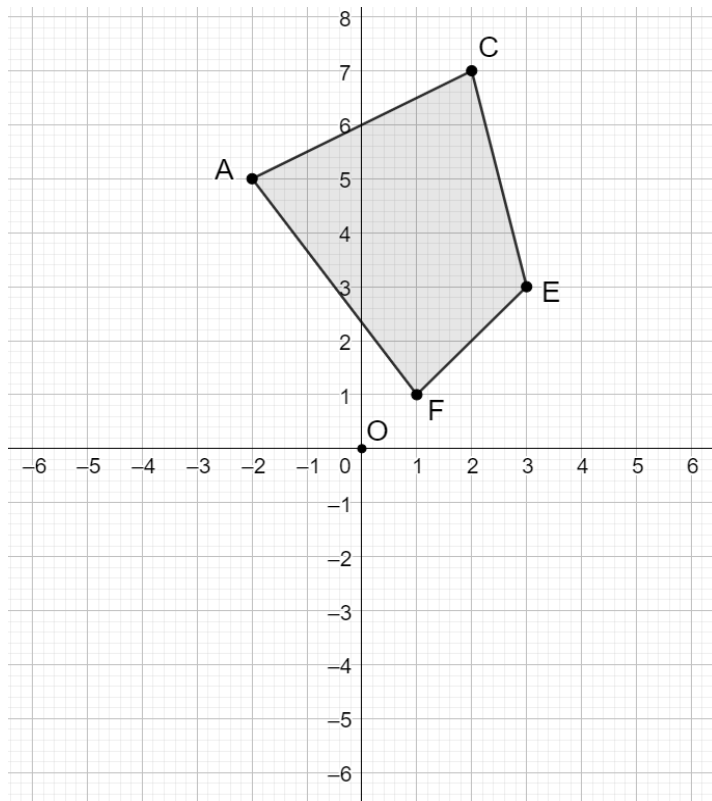
1. Draw the quadrilateral that is a 180° rotation of $L RTP$ about the origin. Label the quadrilateral $L' R' T' P'$.

2. Draw the quadrilateral that is a dilation of $L' R' T' P'$ with a center of dilation at point L' and a scale factor of 2. Label the quadrilateral $L'' R'' T'' P''$.



$ACEF$ and point O are shown on the coordinate plane.

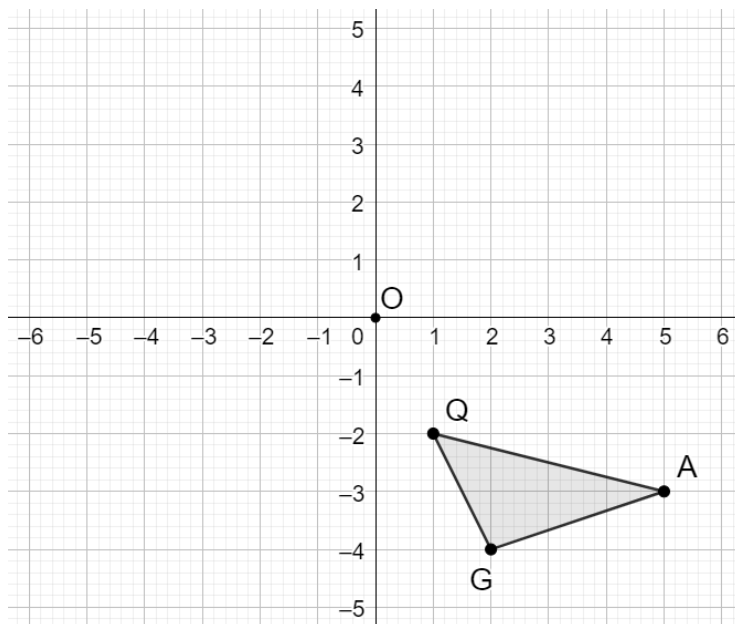
3. Draw the quadrilateral that is a dilation of $ACEF$ with a center of dilation at point O and a scale factor of $\frac{1}{2}$. Label the quadrilateral $A' C' E' F'$.
4. Draw the quadrilateral that is a reflection of $A' C' E' F'$ across the x -axis. Label the quadrilateral $A'' C'' E'' F''$.



$\triangle QAG$ and point O are shown on the coordinate plane.

5. Draw the triangle that is a 90° counterclockwise rotation of $\triangle QAG$ about point O . Label the quadrilateral $\triangle Q'A'G'$.

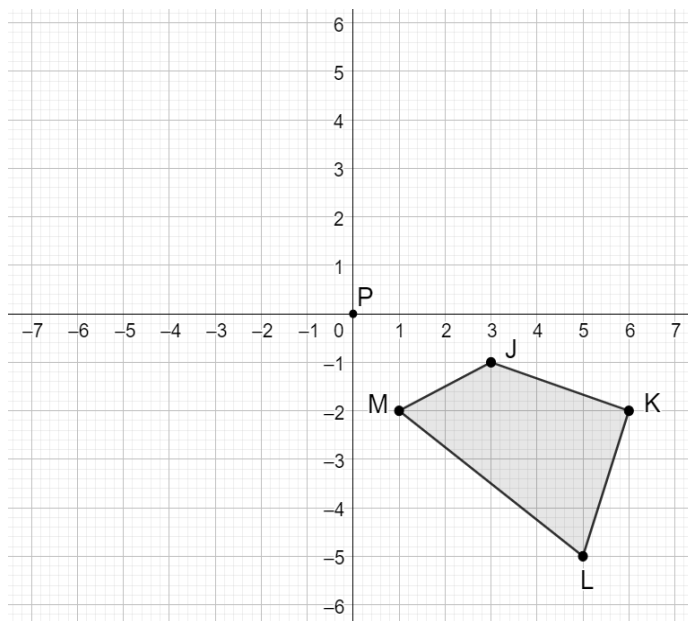
6. Draw the triangle that is a reflection of $\triangle Q'A'G'$ across the y -axis. Label the quadrilateral $\triangle Q''A''G''$.



$JKLM$ and point P are shown on the coordinate plane.

7. Draw the quadrilateral that is a 90° clockwise rotation of $JKLM$ about the origin. Label the quadrilateral $J'K'L'M'$.

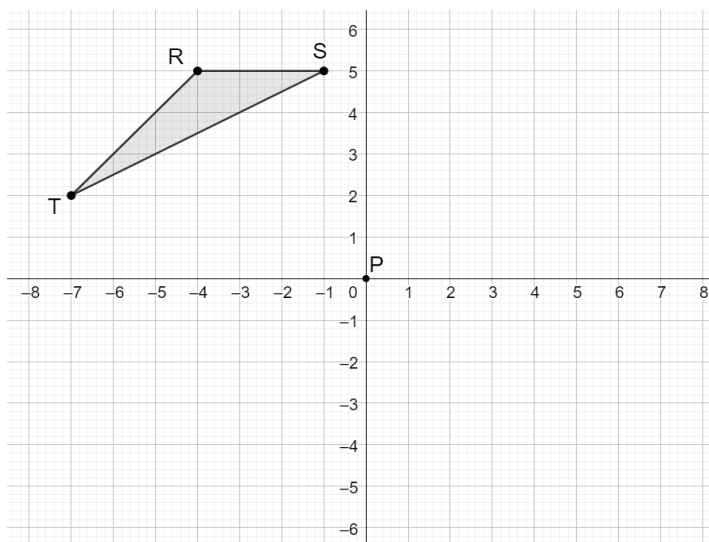
8. Draw the quadrilateral that is a translation of $J'K'L'M'$ 6 units right and 3 units up. Label the quadrilateral $J''K''L''M''$.



$\triangle RST$ and point P are shown on the coordinate plane.

9. Draw $\triangle R'S'T'$ as a result of $(x, y) \rightarrow (x + 7, y - 2)$.

10. Draw $\triangle R''S''T''$ as a result of $(x, y) \rightarrow \left(\frac{2}{3}x, \frac{2}{3}y\right)$.



Geometry
Unit 6
Lesson 6 Practice

Name _____
Date _____
Period _____

1.

- ☐ Only dilations
- ☐ Only translations, reflections, and rotations
- ☐ All transformations

corresponding angles.

result in an image with congruent

- ☐ Only dilations
- ☐ Only translations, reflections, and rotations
- ☐ All transformations

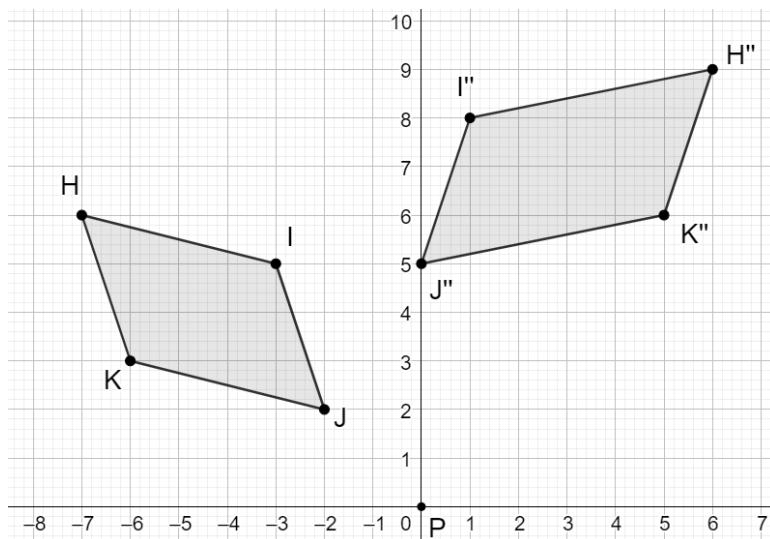
corresponding angles and sides.

result in an image with congruent

- ☐ Only dilations
- ☐ Only translations, reflections, and rotations
- ☐ All transformations

but not congruent corresponding sides.

result in an image with proportional,

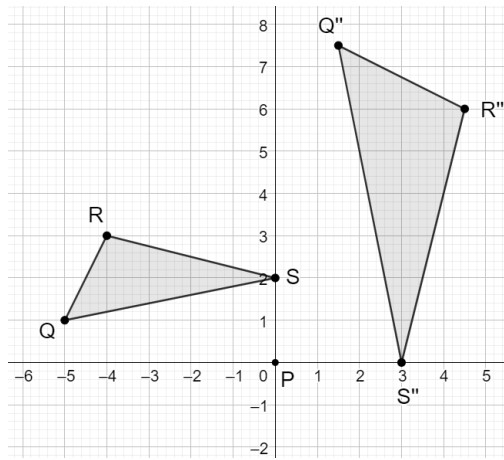


2. Write a sequence of transformations for preimage $HIJK$ that would show whether the figures are similar.

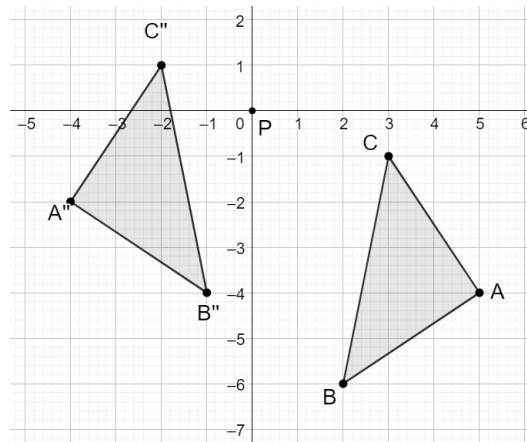
3. Is $H''I''J''K''$ similar to the preimage?

- ☐ Yes
- ☐ No

4. Explain the reason for your answer for #3.



5. Write a sequence of transformations for preimage $\triangle QRS$ that would show whether the figures are similar.
6. Is $\triangle Q''R''S''$ similar to the preimage?
☐ Yes
☐ No
7. Explain the reason for your answer for #6.



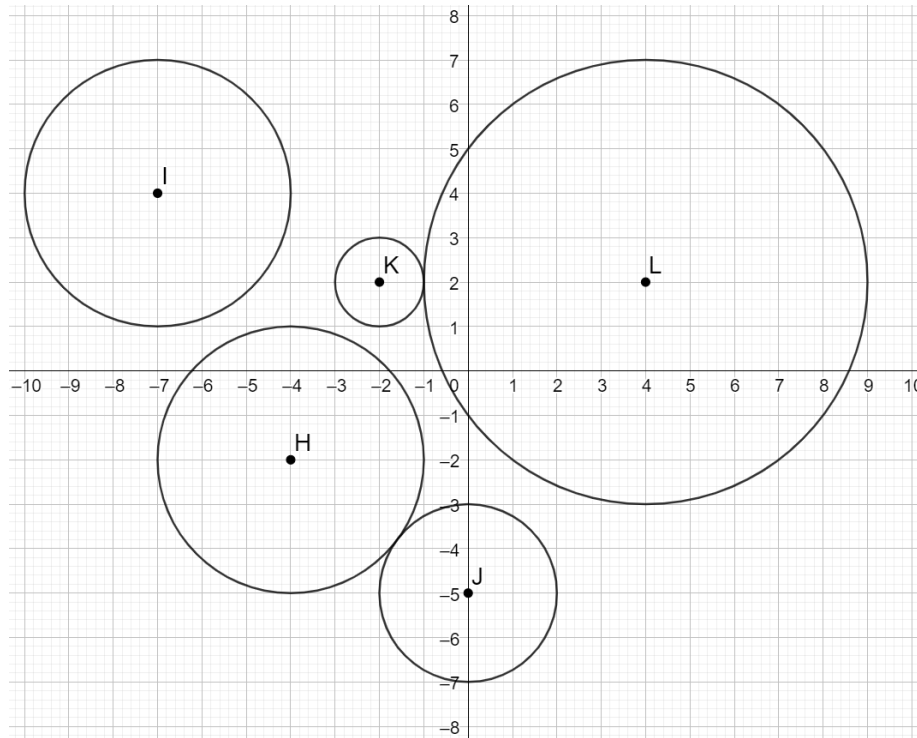
8. Write a sequence of transformations for preimage $\triangle ABC$ that would show whether the figures are similar.
9. Is $\triangle A''B''C''$ similar to the preimage?
☐ Yes
☐ No
10. Explain the reason for your answer for #9.

Geometry
Unit 6
Lesson H1 Practice

Name _____
Date _____
Period _____

1. All circles are

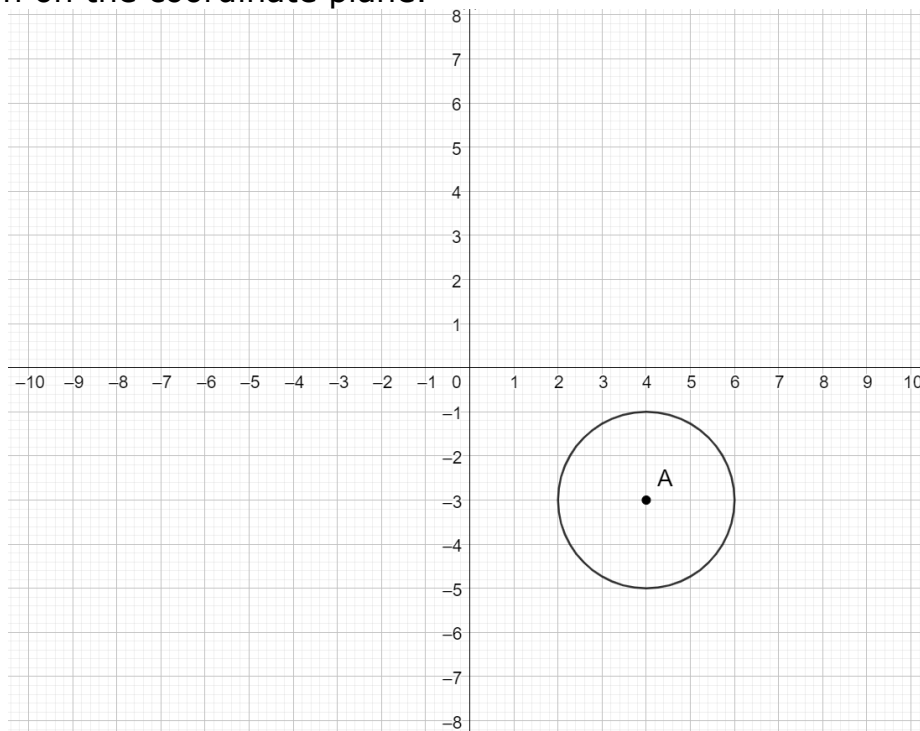
- ☐ congruent
- ☐ similar



Given Circle H as the preimage:

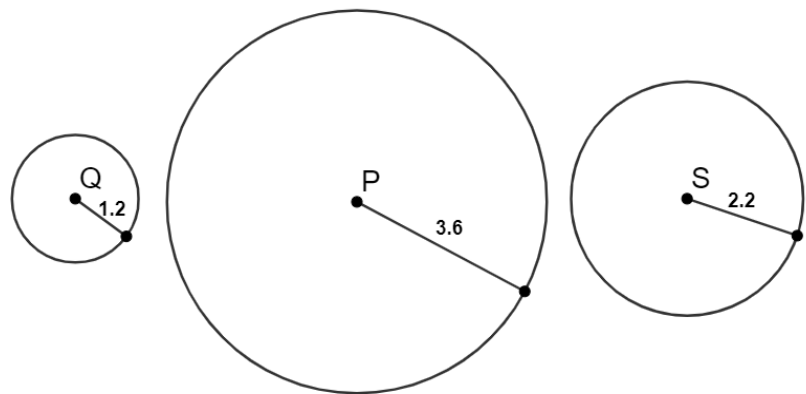
2. How can Circle H be mapped onto Circle I ?
3. How can Circle H be mapped onto Circle J ?
4. How can Circle H be mapped onto Circle K ?
5. How can Circle H be mapped onto Circle L ?

Circle A is shown on the coordinate plane.



6. Draw Circle B , the result of translation 9 units left and 2 units down and a dilation of $\frac{1}{2}$ about the center of Circle A .
7. Draw Circle C , the result of translation 2 units left and 3 units up.
8. Draw Circle D , the result of a dilation of $\frac{3}{2}$ about the center of Circle A and a translation 8 units up.
9. Draw Circle E , the result of translation 4 units right and 2 units down and a dilation of $\frac{1}{2}$ about the center of Circle A .
10. Draw Circle F , the result of translation 6 units left and 7 units up and a dilation of 2 about the center of Circle A .

Circles Q , P , and S are shown. The radius of Circle Q is 1.2 units long. The radius of Circle P is 3.6 units long. The radius of Circle S is 2.2 units long.



1. Complete the following table. Round to the nearest thousandth. Use the π button on your calculator.

Circle	Radius	Circumference

2. Complete the table.

Circle Q and P	
Ratio of Radius of Circle Q and Radius of Circle P	
Ratio of Circumference of Circle Q and Circumference of Circle P	

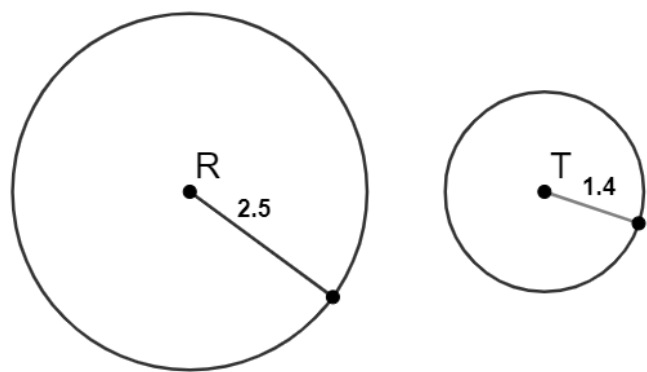
3. Complete the table.

Circle P and S	
Ratio of Radius of Circle P to Radius of Circle S	
Ratio of Circumference of Circle P to Circumference of Circle S	

4. Complete the table.

Circle Q and S	
Ratio of Radius of Circle Q to Radius of Circle S	
Ratio of Circumference of Circle Q to Circumference of Circle S	

Circles R and T are shown. The radius of Circle R is 2.5 units long. The radius of Circle T is 1.4 units long.



Show Circle R is similar to Circle T by completing the proof by contradiction:

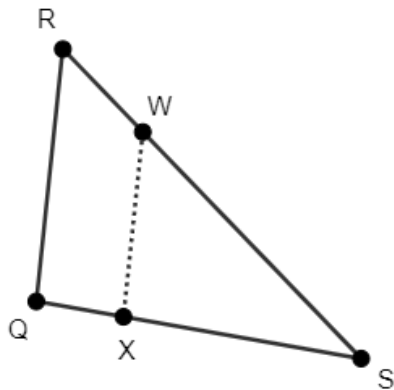
5. Assume that _____ is not similar to Circle T .
6. The ratio of the radius of Circle R to the radius of Circle T is _____.
7. The ratio of the circumference of Circle R to the circumference of Circle T is _____.
8. The ratio of the _____ and the ratio of the _____ are equal, so corresponding parts of Circle R and Circle T are proportional.
9. Since the corresponding parts of Circle R and Circle T are proportional, the circles are similar. This is a contradiction, so the assumption that Circle R is _____ to Circle T is not true.
10. All circles are

congruent

similar

.

$\triangle QRS$ is shown, where $m\angle QRS = 50^\circ$, $m\angle RQS = 94^\circ$, and $m\angle RSQ = 36^\circ$. Point W is on $\overline{RS}$ and Point X is on $\overline{QS}$. $\overline{WX}$ is drawn as a dotted line.



1. Use a ruler and a protractor to complete the table.

Measurements of $\triangle QRS$		Measurements of $\triangle XWS$	
$QR =$	$m\angle QRS = 50^\circ$	$XW =$	$m\angle XWS =$
$RS =$	$m\angle RQS = 94^\circ$	$WS =$	$m\angle WXS = 94^\circ$
$QS =$	$m\angle RSQ = 36^\circ$	$XS =$	$m\angle WSX =$

2. The corresponding angles of $\triangle QRS$ and $\triangle XWS$ are _____.

3. Complete the table.

Sides	Ratio
Determine the ratio of $\overline{QR}$ to $\overline{XW}$.	
Determine the ratio of $\overline{RS}$ to $\overline{WS}$.	
Determine the ratio of $\overline{QS}$ to $\overline{XS}$.	

4. The corresponding sides of $\triangle QRS$ and $\triangle XWS$ are _____.

5. $\triangle QRS$ is similar to $\triangle$ _____.
6. Describe a sequence of transformations that can be used to map $\triangle QRS$ onto $\triangle XWS$.

Complete each relationship for $\triangle QRS$ and $\triangle XWS$.

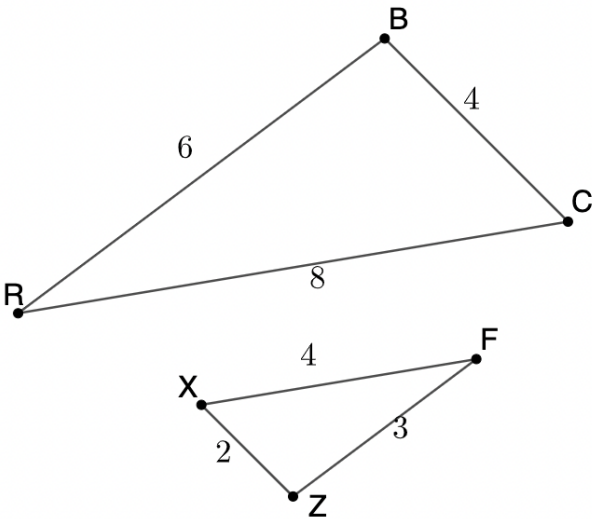
7. $\frac{RS}{\quad} = \frac{\quad}{XS}$

8. $\frac{WX}{\quad} = \frac{\quad}{QS}$

9. $\frac{RQ}{\quad} = \frac{\quad}{WS}$

10. Observing that $\frac{RQ}{WX} \neq \frac{RW}{WS}$, your classmate Charlee claims that $\triangle QRS$ and $\triangle XWS$ are not similar. If her logic is faulty, explain why.

$\triangle RBC$ and $\triangle XFZ$ are shown.



1. Complete the table.

Measurements of $\triangle RBC$	Measurements of $\triangle XFZ$	Ratio
$RB =$	$ZF =$	$\frac{RB}{ZF} =$
$BC =$	$XZ =$	$\frac{BC}{XZ} =$
$CR =$	$XF =$	$\frac{CR}{XF} =$

2. What similarity theorem can be applied to show that the triangles are similar?

3. Write the similarity statement.

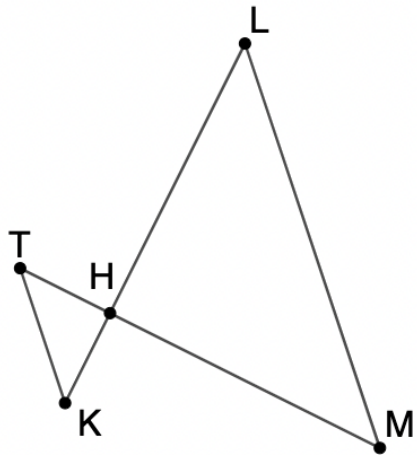
4. Complete each.

$\frac{RC}{XZ} =$

$\frac{XF}{RB} =$

$\frac{BC}{FZ} =$

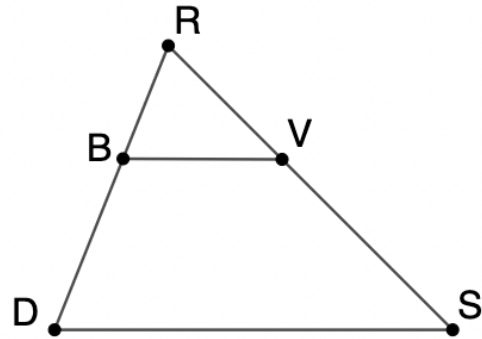
$\triangle TKH$ and $\triangle HLM$ are shown where $\overline{TK} \parallel \overline{LM}$.



Complete the two-column proof.

	Statement	Reason
	$\overline{TK} \parallel \overline{LM}$	Given
5.	$\angle HLM \cong \angle HKT$ and $\angle HML \cong \angle HTK$	
6.	$\triangle TKH \sim \triangle \underline{\hspace{1cm}}$	

$\triangle DRS$ and $\triangle RVB$ are shown.



Given: $RB = \frac{2}{5}RD$ and $RV = \frac{2}{5}RS$
Prove: $\triangle DRS \sim \triangle BRV$

	Statement	Reason
	$RB = \frac{2}{5}RD$ and $RV = \frac{2}{5}RS$	Given
7.	$\frac{RB}{RD} = \frac{2}{5}$ and $\frac{RV}{RS} = \frac{2}{5}$	Division Property of Equality
8.	$\frac{RB}{RD} = \frac{RV}{RS}$	
9.	$\angle DRS \cong \angle BRV$	
10.	$\triangle DRS \sim \triangle BRV$	

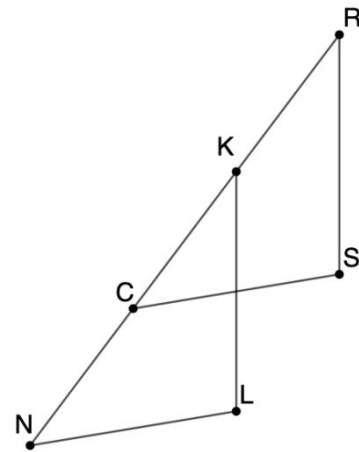
Geometry
Unit 6
Lesson 9 Practice

Name _____
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Period _____

Given: $\overline{NL} \parallel \overline{CS}$ and $\angle KRS$ and $\angle LKR$ are supplementary

Prove: $\triangle NKL \sim \triangle CRS$

Complete the two-column proof with the provided statements and reasons.



	Statement	Reason
	$\overline{NL} \parallel \overline{CS}$	Given
1.		
2.		
3.		
4.		
5.		

$\triangle NKL \sim \triangle CRS$

AA ~

$\angle KRS$ and $\angle LKR$ are supplementary angles

If two parallel lines are intersected by a transversal, then corresponding angles are congruent.

$\angle NKL \cong \angle CRS$

If consecutive interior angles are supplementary, then two lines intersected by a transversal are parallel.

$\overline{KL} \parallel \overline{RS}$

Reflexive Property of Congruence

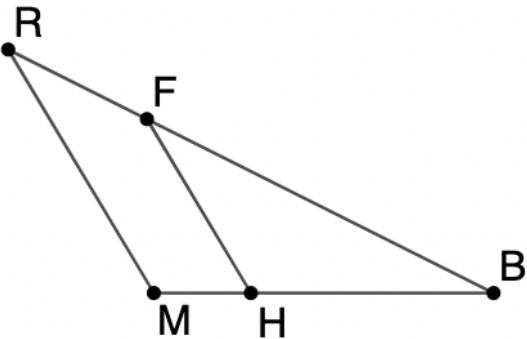
$\angle KNL \cong \angle RCS$

Given

Given: $RF = \frac{2}{7}RB$ and $MH = \frac{2}{7}MB$

Prove: $\triangle RBM \sim \triangle FBH$

Complete the two-column proof with the provided statements and reasons.



	Statement	Reason
6.		
7.		
8.		
9.		
10.		

$\frac{RF}{RB} = \frac{2}{7}$ and $\frac{MH}{MB} = \frac{2}{7}$

$\triangle RBM \sim \triangle FBH$

$\frac{RF}{RB} = \frac{MH}{MB}$

$RF = \frac{2}{7}RB$ and $MH = \frac{2}{7}MB$

$\angle RBM \cong \angle FBH$

SAS ~

Given

Reflexive Property of Congruence

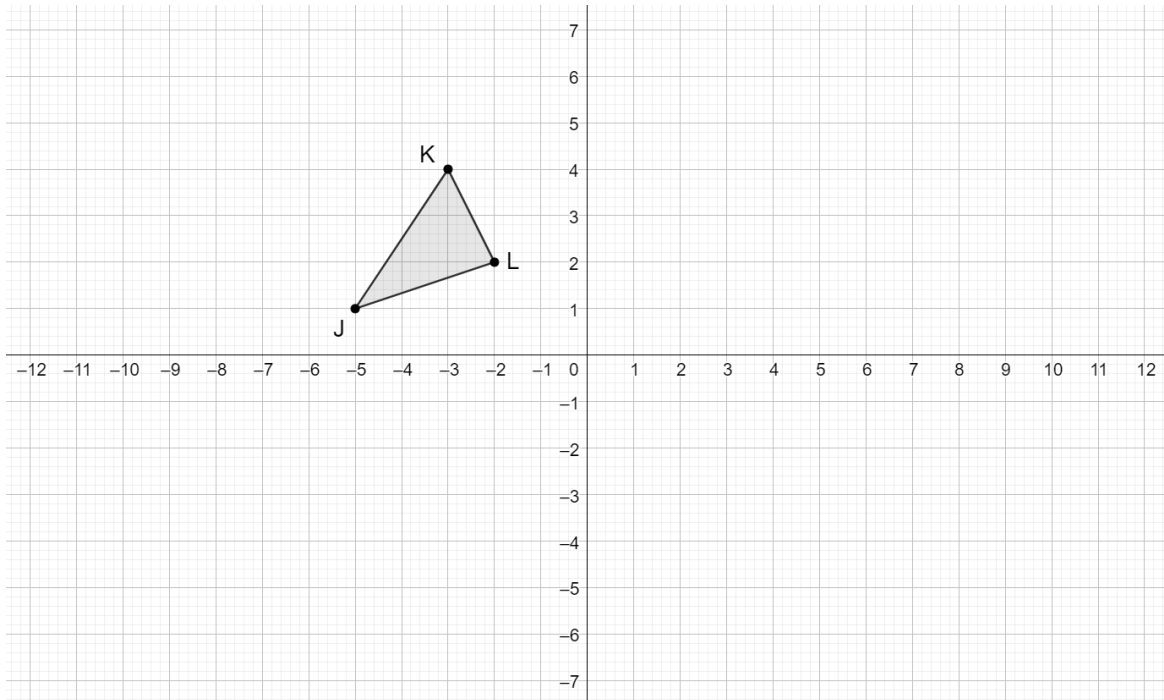
Substitution Property of Equality

Division Property of Equality

Geometry
Unit 6
Lesson H3 Practice

Name _____
Date _____
Period _____

$\triangle JKL$ is shown.



Draw the result of each transformation in the following sequence:

1. $\triangle J'K'L'$ as a result of $(x, y) \rightarrow (y, -x)$
2. $\triangle J''K''L''$ as a result of $(x, y) \rightarrow \left(\frac{3}{2}x, \frac{3}{2}y\right)$
3. $\triangle J'''K'''L'''$ as a result of $(x, y) \rightarrow (x, -y)$

Indicate the correct relationship for the resulting image:

4. Image $\triangle J'K'L'$ is

☐ congruent
☐ similar

 to preimage $\triangle JKL$.
5. Image $\triangle J''K''L''$ is

☐ congruent
☐ similar

 to preimage $\triangle JKL$.
6. Image $\triangle J'''K'''L'''$ is

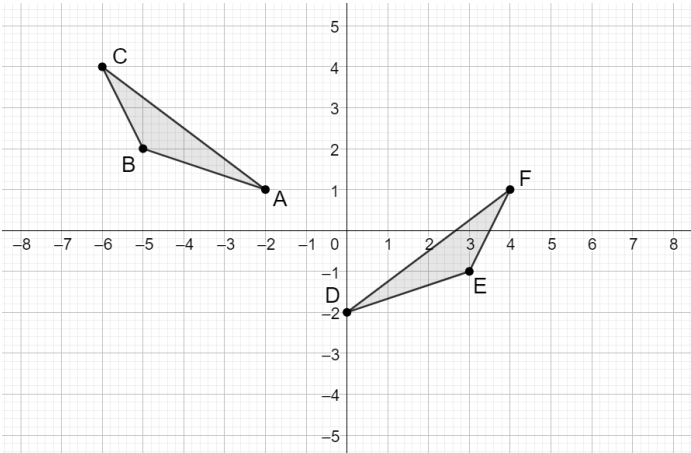
☐ congruent
☐ similar

 to preimage $\triangle JKL$.

$\triangle ABC$ and $\triangle DEF$ are shown.

Assume: $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$

7. Write a series of transformations that will map $\triangle ABC$ onto $\triangle DEF$ so that $\triangle ABD \sim \triangle DEF$.



8. If it is given that $\triangle DEF$ is a result of $(x,y) \rightarrow (-x,y)$ then $(x,y) \rightarrow (x-2,y-3)$, then the triangles are

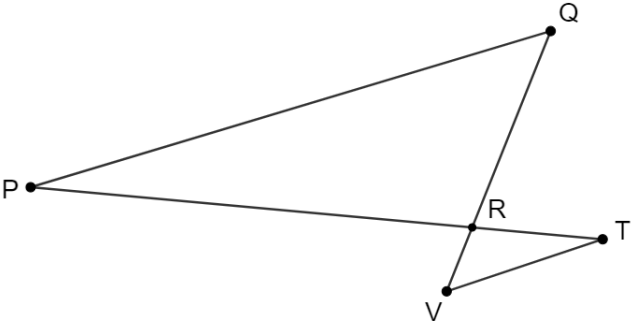
congruent

similar

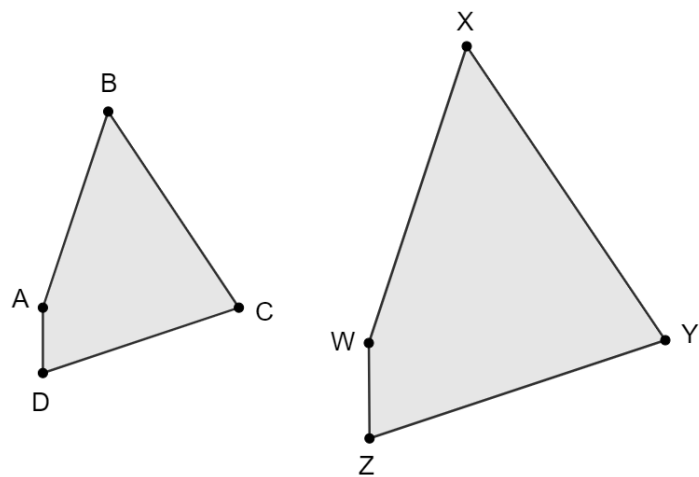
.

$\triangle PQR$ and $\triangle TVR$ are shown.

Complete the table:



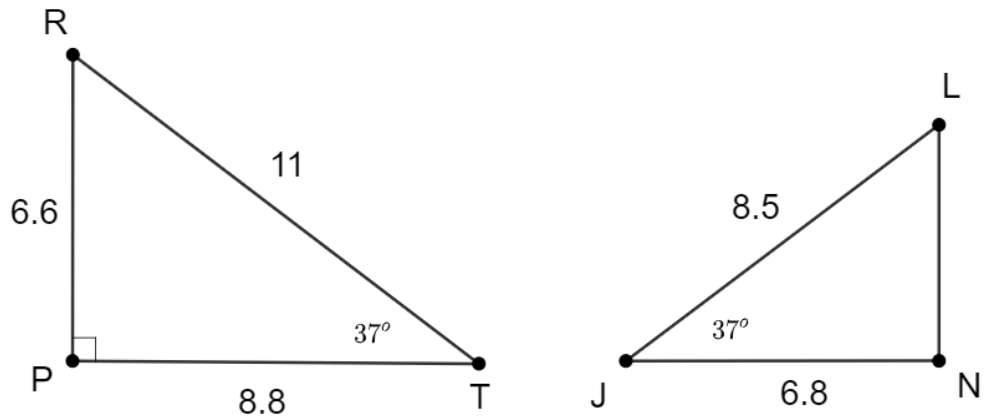
	If Given	True or False? $\triangle PQR \sim \triangle TVR$ by AA Similarity	Justification
9.	$\frac{TR}{PR} = \frac{VR}{QR}$	True False	
10.	$\overline{PQ} \parallel \overline{VT}$	True False	



Fill in the table given that $ABCD \sim WXYZ$.

	Corresponding Angles	Corresponding Sides
1.	$\angle DAB \cong \angle$	$\frac{AB}{\quad} = \frac{DC}{\quad}$
2.	$\angle WZY \cong \angle$	$\frac{WZ}{\quad} = \frac{XY}{\quad}$
3.	$\angle XYZ \cong \angle$	$\frac{BC}{\quad} = \frac{WX}{\quad}$
4.	$\angle ABC \cong \angle$	$\frac{ZY}{\quad} = \frac{AD}{\quad}$

5. If $WZ = 3$, $XY = 10.5$, and $AD = 2$, determine $\overline{BC}$.



$\triangle PRT$ and $\triangle JLN$ are shown. $PR = 6.6$, $PT = 8.8$, $RT = 11$, $JL = 8.5$, $JN = 6.8$, $\overline{RP} \perp \overline{PT}$, $m\angle PTR = 37^\circ$, and $m\angle LJN = 37^\circ$.

6. What similarity theorem or postulate could be used to prove the triangles are similar?

7. Write a similarity statement for $\triangle PRT$ and $\triangle JLN$.

Complete the table.

	Angle Measures	Side Proportion or Length
8.	$m\angle PRT =$	$\frac{RT}{LJ} =$
9.	$m\angle JLN =$	$\frac{PT}{JN} =$
10.	$m\angle LNJ =$	$LN =$

Geometry
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$DKLM$ and $PTRS$ are shown, where $DKLM \sim PTRS$ with $MD = 21$, $DK = 4y - 2$, $KL = 6x + 3$, $LM = 36$, $PT = 10$, $TR = 11$, and $RS = 12$.

1. Determine the value of each variable.

$$x = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{2cm}}$$

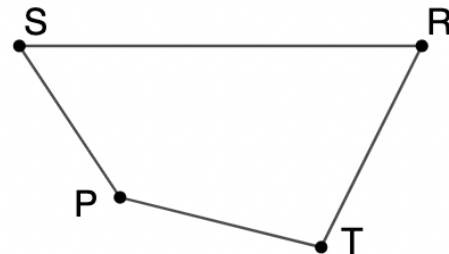
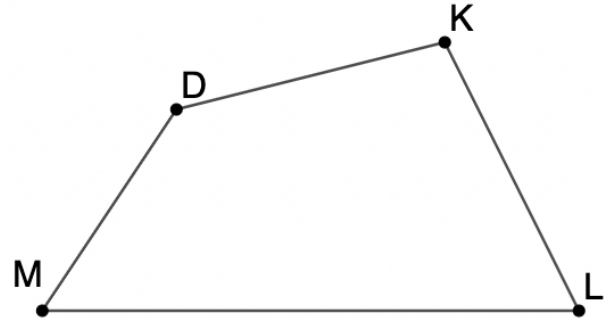
2. Determine the value of each side.

$$DK = \underline{\hspace{2cm}}$$

$$SP = \underline{\hspace{2cm}}$$

$$KL = \underline{\hspace{2cm}}$$

3. What is the perimeter of $DKLM$?



Consider additional measurements of angles for $DKLM$ and $PTRS$ where $m\angle MDK = (12x + 15)^\circ$, $m\angle SPT = (14x - 5)^\circ$, $m\angle DKL = 115^\circ$, $m\angle DML = (8y + 12)^\circ$, $m\angle PSR = (13y - 18)^\circ$.

4. Determine the value of each variable.

$$x = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{2cm}}$$

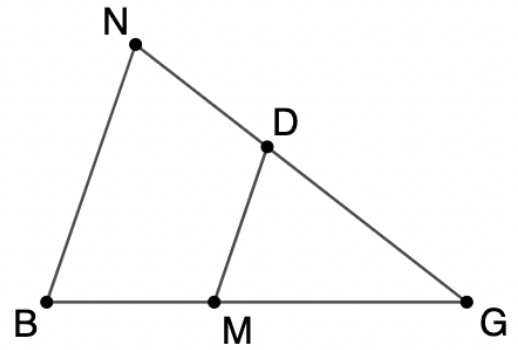
5. Determine the value of each angle.

$$m\angle MDK = \underline{\hspace{2cm}}$$

$$m\angle PTR = \underline{\hspace{2cm}}$$

$$m\angle PSR = \underline{\hspace{2cm}}$$

$\triangle NBG$ and $\triangle DMG$ are shown, where $\triangle NBG \sim \triangle DMG$ with $DM = 6$, $NB = 21$, $DG = 4$, $MG = 2x + 3$, and $BG = 13x - 1.5$.



6. What is the scale factor from $\triangle DMG$ to $\triangle NBG$?

7. What is the value of the variable, x ?

8. What is the length of $\overline{BG}$?

Consider additional measurements of angles for $\triangle NBG$ and $\triangle DMG$ where $m\angle GNB = (20y + 10)^\circ$, $m\angle GMD = (5x - 5)^\circ$, $m\angle GBN = (4x + 10)^\circ$, $m\angle GDM = (25y - 5)^\circ$.

9. Determine the value of each variable.

$$x = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{2cm}}$$

10. Determine the value of each angle.

$$m\angle GNB = \underline{\hspace{2cm}}$$

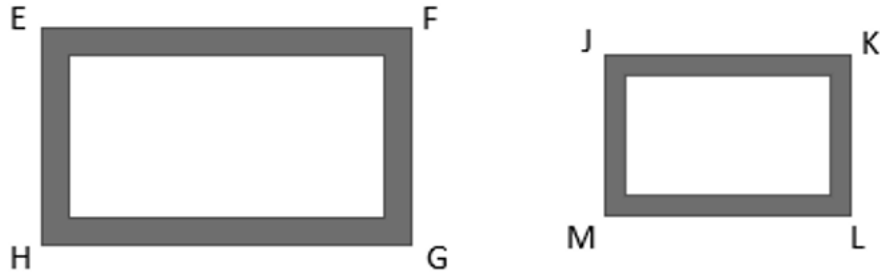
$$m\angle GBN = \underline{\hspace{2cm}}$$

$$m\angle DGM = \underline{\hspace{2cm}}$$

Geometry
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Tamika is custom building frames for a set of family photos. She has designed two similar rectangular frames, $EFGH$ and $JKLM$. The diagram shows the two different sized frames, where $JM = 6$ inches (in.), $EF = (3w + 2)$ in., $FG = 10$ in., and $JK = (2w + 0.8)$ in.



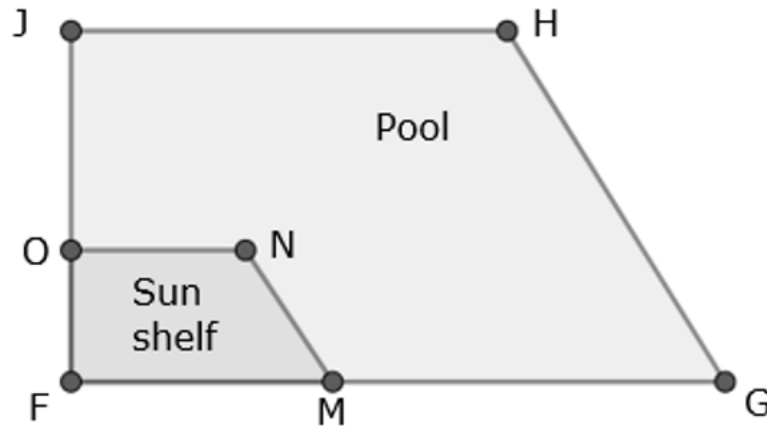
1. What is the value of w ?
2. What is the area of $EFGH$? Include units.
3. What is the perimeter of $JKLM$? Include units.

Tamika created a third similar rectangular frame, $PQRS$, shown below, where $PQ = 6$ in.



4. What is the length of $\overline{SP}$? Include units.
5. What is the perimeter, in inches, of $PQRS$?

Shelley and her family are having a pool built in the backyard. The pool, $FGHJ$, is shaped like a trapezoid, with a smaller similar trapezoid, $FMNO$, for a sun shelf. A diagram of the pool is shown, where $FG = (12k - 6)$ feet (ft), $FM = 10.5$ ft, $NM = 12$ ft, $HG = 24$ ft, $HJ = (12m - 2)$ ft, and $NO = (4m + 3)$ ft.



6. Determine the value of each variable:

$$k = \underline{\hspace{2cm}}$$

$$m = \underline{\hspace{2cm}}$$

7. Determine the value of each side:

$$HJ = \underline{\hspace{2cm}}$$

$$FG = \underline{\hspace{2cm}}$$

8. If $HJ = JF$, what is the perimeter, in feet, of the pool?

Some measurements of angles of $FGHJ$ and $FMNO$ are given.

$$m\angle FMN = (15y + 10)^\circ, m\angle FGH = (18y - 2)^\circ, m\angle OFM = 90^\circ,$$

$$m\angle FJH = (5w + 15)^\circ, m\angle FON = (15w - 15)^\circ$$

9. Determine the value of each variable:

$$y = \underline{\hspace{2cm}}$$

$$w = \underline{\hspace{2cm}}$$

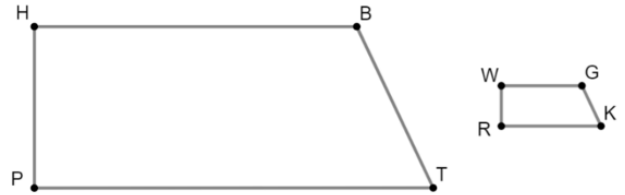
10. What is the measure of $\angle ONM$?

Geometry
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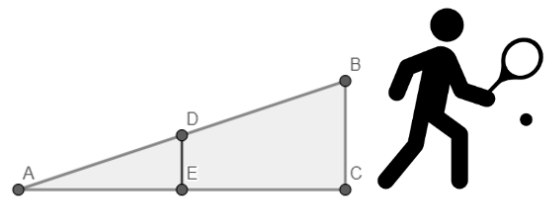
Wendy, a party planner, is designing a banner for a retirement party. Her client wants two similar banners: a large banner, $HBTP$, for the main entrance, and a similar sized banner, $WGKR$, for a photo booth.

Wendy knows that $PT = 72$ inches (in),
 $HP = (5x - 3)$ in., $WG = 6$ in., $WR = (3x - 10)$ in.,
 $BT = 40$ in., and $RK = 9$ in.



1. What scale factor will map $WGKR$ to $HBTP$?
2. What is the perimeter of $HBTP$, the main entrance banner?
3. What is the perimeter of $WGKR$, the photo booth banner?

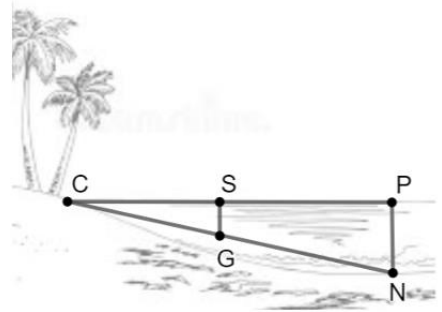
In the diagram, $\triangle ADE \sim \triangle ABC$, $AE = 7$ meters (m),
 $DE = 0.93$ m, and $EC = 21$ m. Serena is doing a
tennis drill where she wants to hit the ball so that
it will just pass over the net, $\overline{DE}$, and land 7 m
away from the base of the net.



4. What scale factor will map $\triangle ABC$ to $\triangle ADE$?
5. What is the height, represented by $\overline{BC}$, of the tennis racket?

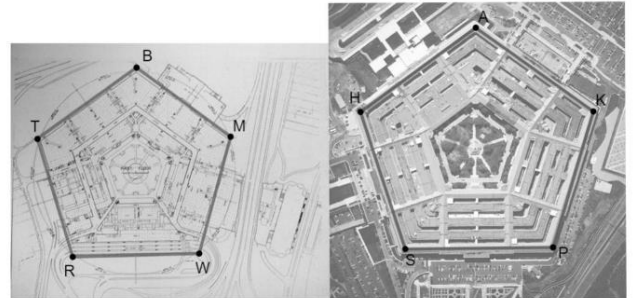
A lake front property owner is building a dock which extends the length of $\overline{CP}$ into the lake. $\overline{PN}$ and $\overline{SP}$ are the two main support posts for the dock.

$\triangle CSG \sim \triangle CPN$, $CS = 6.5$ feet (ft), $SG = (2x - 1)$ ft,
 $PN = (3x + 9)$ ft, and $CP = 32.5$ ft.



6. What scale factor will map $\triangle CPN$ to $\triangle CSG$?
7. What is the depth of the longer support post, represented by $\overline{PN}$?
8. What is the depth of the shorter support post, represented by $\overline{SG}$?

Cesar wants to create a scale model of the Pentagon, where all sides are equal, for his social studies project. A copy of the blueprint, $BMWRT$, will guide him in building a similar scale model, $AKPSH$. The blueprint's sides, $BMWRT$, are $\frac{1}{6}$ of the scale model Pentagon, $AKPSH$.
 $TB = (0.5x + 5)$ centimeters (cm) and
 $HA = (5x + 10)$ cm.



9. What is the perimeter of the Pentagon in the blueprint, $BMWRT$?
10. What is the perimeter of the scale model Pentagon, $AKPSH$?